

CURRICULUM VITAE (CVA)**Part A. PERSONAL INFORMATION****CV date**

11/03/2025

First name	Pablo		
Family name	Catalán Fernández		
Gender (*)	Male	Birth date (dd/mm/yyyy)	28/02/1988
ID number	51472311J		
e-mail	pcatalan@math.uc3m.es	URL Web	pablocatalan.github.io
Open Researcher and Contributor ID (ORCID) (*)	0000-0003-2826-4684		

(*) Mandatory

A.1. Current position

Position	Profesor Titular de Universidad (Associate Professor)		
Starting date (dd/mm/yyyy)	04/03/2025		
Institution	Universidad Carlos III de Madrid		
Department/Center	Mathematics		
Country	Spain	Phone	0034645553887
Key words	systems biology, antibiotic resistance, evolutionary biology, mathematical biology, complex systems		

Member of the [Grupo Interdisciplinar de Sistemas Complejos \(GISC\)](#)**A.2. Previous positions**

Period	Position/Institution/Country
Sep 2019- Mar 2025	Assistant Professor, Universidad Carlos III de Madrid
Sep 2018-Sep 2019	Ramón Areces Postdoctoral Fellow, University of Exeter (UK)
Apr 2017-Jun 2018	Postdoctoral Researcher, Universidad Carlos III de Madrid
Dec 2012-Mar 2017	FPI Predoc Fellow, Universidad Carlos III de Madrid

A.3. Education

PhD, Licensed, Graduate	University/Country	Year
BSc in Mathematics	Universidad Nacional de Educación a Distancia	2017
PhD in Mathematical Engineering	Universidad Carlos III de Madrid	2017
Msc in Modeling and Physics of Complex Systems	Universidad Rey Juan Carlos	2012
BSc + MSc in Biology	Universidad Complutense de Madrid	2011

Part B. CV SUMMARY (max. 5000 characters, including spaces)

Pablo Catalán is a multidisciplinary researcher with expertise in Biology, Physics, and Mathematics, integrating theoretical modeling with experimental biology. Since March 2025, he is an Associate Professor at Universidad Carlos III de Madrid (UC3M), where he carries out research and teaches mathematics courses to engineering students.

Prof. Catalán earned his Ph.D. in Mathematical Engineering from UC3M with honors (sobresaliente cum laude), specializing in genotype-phenotype maps and evolutionary dynamics through computational models. During his doctoral studies, he developed toyLIFE, a computational model of simplified cellular biology that deepens our understanding of molecular evolution and genotype-phenotype relationships. His Ph.D. work was conducted under the supervision of Prof. José A. Cuesta, in close collaboration with Prof. Susanna Manrubia at the Centro Nacional de Biotecnología (CNB, CSIC), and resulted in nine publications, including a recent high-impact review.

After completing his Ph.D., Prof. Catalán was awarded a Ramón Areces Postdoctoral Fellowship, which allowed him to join the University of Exeter as a researcher with Prof.



Robert Beardmore. There, he worked on mathematical models of antibiotic resistance evolution and treatment strategies, leading to high-impact publications, including a paper in *Nature Communications*. He also contributed to the analysis of ATLAS, one of the largest antibiotic resistance databases, which earned him the Innovation Award in Vivli's 2023 AMR Data Challenge.

In 2019, he returned to UC3M as an Assistant Professor (Profesor Ayudante Doctor), rejecting a Juan de la Cierva fellowship. Over the past years, he has expanded his work to include applied experimental microbiology, particularly through a collaboration with Prof. Saúl Ares (CNB, CSIC). This collaboration, initially focused on systems biology in plants, resulted in a 2022 *Science Advances* publication, modeling the molecular network governing plant responses to light and temperature. More recently, they have shifted their focus to antibiotic resistance, co-leading the establishment of an experimental microbiology lab to test predictions from mathematical models of bacterial adaptation and drug resistance. This research has been supported by two major grants from the Spanish Research Agency, the most recent of which was obtained in 2023 with Prof. Catalán as co-Principal Investigator.

Prof. Catalán is particularly interested in sequential antibiotic therapies, an approach that exploits evolutionary trade-offs to optimize treatment strategies. His research in this area integrates machine learning, AI-driven analysis, and experimental validation to identify effective therapy sequences that minimize resistance evolution. His team at UC3M includes Javier Molina and Lucía Yubero, two Ph.D. students co-supervised with Prof. Ares.

Beyond his work in antibiotic resistance, Prof. Catalán has recently expanded his research collaborations internationally. In partnership with Dr. Yi Jin (University of Manchester), he is developing models of enzyme kinetics relevant to antibiotic resistance, resulting in a recent publication in *ACS Central Science*. Additionally, he has initiated a collaboration with Prof. Álvaro Sánchez (Instituto de Biología Funcional y Genómica, CSIC-Salamanca) to study microbial interactions and evolutionary dynamics, leading to a paper in *eLife*.

Pablo has presented at numerous national and international conferences, delivering 10 talks (including four invited lectures) and ten poster presentations. He holds one six-year research period recognition (sexenio).

In addition to his research, Prof. Catalán has over 700 hours of teaching courses such as Calculus I and II, Linear Algebra, Advanced Mathematics or Mathematics for Neuroscience (all teaching done in English). His teaching is very well valued by students, with an average mark of 4.44 (out of 5) in student surveys. He has also developed educational content, including an online Calculus course available through OCW-UC3M and YouTube. He holds one five-year teaching period recognition (quinquenio).

Prof. Catalán is deeply committed to scientific outreach and education. He has participated in multiple STEM initiatives, including "4º ESO y Empresa," "STEM Fridays," "STEM for Girls," and the Concurso de Matemática Aplicada y Computacional at UC3M. He has also mentored six master's students and supervised six undergraduate theses, contributing to the development of young researchers in his field.

Prof. Catalán's work is characterized by a strong interdisciplinary approach, combining mathematical models, high-throughput data analysis, AI tools, and experimental research. His GitHub repository contains software developed during his projects, ensuring transparency and accessibility of his research tools. His long-term goal is to bridge the gap between theory and practice in antibiotic resistance research, developing innovative strategies that can have a real-world impact on infectious disease treatment.

Part C. RELEVANT MERITS (*sorted by typology*)

C.1. Publications



Selected publications from the last 5 years. The full list can be consulted at [Pablo Catalán's Google Scholar profile](#).

[An asterisk indicates shared co-authorship. The journal's JCR Impact Factor and its ranking are indicated in brackets.]

1. Díaz-Colunga, J.* , **Catalán, P.***, San Román, M.* , Arrabal, A.* and Sánchez, A. 2024. [Full factorial construction of synthetic microbial communities](#). *eLife* 13, RP101906 [IF: 6.4 (2023), R: 8/109 (Q1, T1, D1) (BIOLOGY)]
2. Rodríguez-Maroto, G.* , **Catalán, P.***, Nieto, C., Prat, S. and Ares, S. 2024. [Mathematical Modeling of Photo- and Thermomorphogenesis in Plants](#). En: Chen, M. (Ed.), *Thermomorphogenesis. Methods in Molecular Biology*, vol. 2795 (capítulo 23). Humana, New York, NY. ISBN: 978-1-0716-3813-2 (book chapter)
3. Zhou, Qi* , **Catalán, P.***, Bell, H.* , Baumann, P., Evans, R., Yang, J., Zhang, Z., Zappala, D., Zhang, Y., Blackburn, G. M., He, Y. and Jin, Y. 2023. [An Ion-Pair Induced Intermediate Complex Captured in Class D Carbapenemase Reveals Chloride Ion as a Janus Effector Modulating Activity](#). *ACS Central Science* 9, 2339. [IF: 12.7, R: 24/230 (Q1, T1) (CHEMISTRY, MULTIDISCIPLINARY)]
4. **Catalán, P.**, García-Martín, J. A., Aguirre, J., Cuesta, J. A. and Manrubia, S. 2023. [Entropic contribution to phenotype fitness](#). *Journal of Physics A: Mathematical and Theoretical* 56, 345601. [IF: 2.0, R:18/60 (Q2, T1) (PHYSICS, MATHEMATICAL), 47/110 (Q2, T2) (PHYSICS, MULTIDISCIPLINARY)]
5. Nieto, C.* , **Catalán, P.***, Luengo, L. M., Legris, M., López-Salmerón, V., Davière, J. M., Casal, J. J., Ares, S. and Prat, S. 2022. [COP1 dynamics integrate conflicting seasonal light and thermal cues in the control of Arabidopsis elongation](#). *Science Advances* 8:eabp8412 [IF: 13.6, R: 7/73 (Q1, T1, D1) (MULTIDISCIPLINARY SCIENCES)]
6. **Catalán, P.**, Wood, E., Blair, J. M. A., Gudelj, I., Iredell, J. R. and Beardmore, R. E. 2022. [Seeking patterns of antibiotic resistance in ATLAS, an open, raw MIC database with patient metadata](#). *Nature Communications* 13:2917 [IF: 16.6, R: 6/73 (Q1, T1, D1) (MULTIDISCIPLINARY SCIENCES)]
7. Manrubia, S., Cuesta, J.A., Aguirre, J., Ahnert, S.E., Altenberg, L., Cano, A.V., **Catalán, P.**, Diaz-Uriarte, R., Elena, S.F., García-Martín, J.A., Hogeweg, P., Khatri, B.S., Krug, J., Louis, A.A., Martin, N.S., Payne, J.L., Tarnowski, M.J. and Weiß, M. 2021. [From genotypes to organisms: State-of-the-art and perspectives of a cornerstone in evolutionary dynamics](#). *Physics of Life Review* 38: 55-106 [IF: 9.833, R: 5/94 (Q1, T1, D1) (BIOLOGY), 5/72 (Q1, T1, D1) (BIOPHYSICS)] (Salvo los dos primeros autores, el resto siguen orden alfabético)
8. Reding, C., **Catalán, P.**, Jansen, G., Bergmiller, T., Wood, E., Rosenstiel, P., Schulenburg, H., Gudelj, I. and Beardmore, R. 2021. [The Antibiotic Dosage of Fastest Resistance Evolution: gene amplifications underpinning the inverted-U](#). *Molecular Biology and Evolution* 38:3847-3863. [IF: 8.8, R: 5/51 (Q1, T1, D1) (EVOLUTIONARY BIOLOGY), 15/175 (Q1, T1, D1) (GENETICS & HEREDITY)]
9. **Catalán, P.**, Manrubia, S. and Cuesta, J. A. 2020. [Populations of genetic circuits are unable to find the fittest solution in a multilevel genotype-phenotype map](#). *Journal of the Royal Society Interface* 17:20190843. [IF: 4.118, R: 19/72 (Q2, T1) (MULTIDISCIPLINARY SCIENCES)]

C.2. Conferences and posters

Selected presentations and posters from the last 5 years. The full list [can be consulted here](#).

Talks

- **Catalán, P.** 2022. *COP1 dynamics integrate conflicting seasonal light and thermal cues in the control of Arabidopsis elongation*. IV BioinfoCAM, October 20, 2022, Madrid (Spain).
- **Catalán, P.** 2019. *Phenotypic bias and evolutionary predictability in a pattern-formation genotype-phenotype map*. CECAM Workshop: *From sequences to functions: challenges in the computation of realistic genotype-phenotype maps.*, March 13-15, 2019, Zaragoza (Spain) (*invited talk*).

Posters

- **Catalán, P.**, Rodríguez-Maroto, G., Nieto, C., Prat, S., & Ares, S. 2025. *Integrating conflicting seasonal light and thermal cues in the control of Arabidopsis elongation*. 3rd Meeting of the Spanish Society of Complex Systems CS³. February 19-21, 2025, Madrid (Spain).
- **Catalán, P.**, Manrubia, S., & Cuesta, J. A. 2021. *Populations of genetic circuits are unable to find the fittest solution in a multilevel genotype-phenotype map*. EMBO Workshop *Predicting Evolution.*, June 4-6, 2021, online.

C.3. Research projects and grants

Patterns of growth and evolution (PGE). Coordinated grant funded by the Spanish Research Agency (AEI), from 01/09/2023 to 31/08/2026. Coordinator: Saúl Ares.

PI subgrant 1 (CNB): Saúl Ares (PID2022-142185NB-C21).

PIs subgrant 2 (UC3M): **Pablo Catalán** and Javier Muñoz-García (PID2022-142185NB-C22)
Funding: 150.000 € (CNB), 37.000 € (UC3M).

Physics of Bacterial and Developing Systems (BADs). Grant funded by AEI (PID2019-109320GB-I00) from 01/06/2020 to 29/02/2024.

PIs: Saúl Ares y Javier Muñoz.

Funding: 52.030 € (CNB).

Complejidad Biológica y Social (BASIC). Grant funded by MICIU (PGC2018-098186-B-I00) from 01/01/2019 to 31/12/2022.

PI: José A. Cuesta.

Funding: 123.783 € (UC3M).

Variación, replicación y adaptación en procesos evolutivos (VARIANCE). Grant funded by MINECO (FIS2015-64349-P) de 01/06/2016 a 31/5/2019.

PI: José A. Cuesta.

Funding: 59.290 € (UC3M).

Procesos dinámicos evolutivos: virus, ecosistemas y comportamientos sociales (PRODIEVO). Grant funded by MINECO (FIS2011-22449) de 01/01/2012 a 31/12/2014.

PI: José A. Cuesta. Funding: 39.390 € (UC3M).

C.4. Others

- Pablo has developed several software libraries as a result of his research, which he shares [on his Github page](#).
- Reviewer for JCR journals, including: Nature Communications, Journal of the Royal Society Interface, PloS Computational Biology, Scientific Reports, EPL, Bulletin of Mathematical Biology, Proceedings of the Royal Society B.
- Organizer of joint seminars on evolutionary dynamics, theoretical ecology, and systems biology between UC3M and CNB (2017-present).